Aman Kumar

Software Engineer | **Backend Development** | B.Tech (Computer Science & Engineering), IKGPTU Jalandhar
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EDUCATION

IK Gujral Punjab Technical University | B.Tech (Computer Science & Engineering) | CGPA: **8.52**/10 *Aug 2023 – May 2027*
Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks

TECHNICAL SKILLS

- **Programming Languages:** Java, JavaScript, SQL
- **Backend:** Node.js, Express.js, REST APIs, WebSockets
- **Databases & Tools:** PostgreSQL, MongoDB, MySQL, Git, GitHub, Postman
- **Concepts:** Authentication & Authorization, API Design, Database Design, Caching, Concurrency

PROFESSIONAL EXPERIENCE

Indian Institute of Technology (IIT Ropar), VLED Lab | **Summer Intern** *May 2026 – Jul 2026*
Full-Stack Development (MERN), Machine Learning *Rupnagar, Punjab, India*

- Developed PippaQ, a semantic Q&A platform serving role-based access control, question-paper reputation scoring, and RAG-powered duplicate detection, working across Express.js, MongoDB, and React in a 5-member team. [Docs]
- Diagnosed and resolved a production middleware failure that caused all APIs to return HTTP 500 responses, restoring platform-wide functionality through a merged fix. [PR #2]
- Co-developed Spandan v0.8.0, a real-time classroom polling system using Socket.IO event synchronization, role-based authentication, and Whisper-powered transcription for automated question generation.

Excellence Technology | **Backend Trainee** *Jun 2025 – Jul 2025*
Full-Stack Development *Chandigarh, India*

- Built CodeBlog, a MERN-stack blogging platform implementing authentication, content publishing, and CRUD-based content management workflows. [Live | GitHub]

OPEN SOURCE CONTRIBUTION

ArchScope [GitHub]

- Implemented a dynamic Contributors section by integrating the GitHub API, building a cached backend route, and enabling automatic contributor discovery with a responsive horizontal-scroll UI. [PR #56]
- Developed a time-complexity based latency modeling feature supporting O(1), O(log n), O(n), and O(n²) growth patterns, enabling more realistic API performance benchmarking and simulation. [PR #59]

PROJECTS

LatencyLens | **API Latency Benchmarking Tool** [Live | GitHub]

- Built LatencyLens, an API benchmarking platform that generates and analyzes 10,000 concurrent HTTP requests while computing p50, p90, p95, and p99 latency metrics with under 5ms instrumentation overhead.
- Architected the system using Clean Architecture principles, implementing 12+ REST APIs, Server-Sent Events (SSE) for live benchmark tracking, and MongoDB-backed historical result storage to improve maintainability and observability.
- Developed a time-complexity driven latency simulation engine supporting O(1), O(log n), O(n), and O(n²) growth models, enabling realistic workload simulation for performance analysis.

Resume Roaster | **AI Resume ATS Checker** [Live | GitHub]

- Built Resume Roaster, an AI-powered resume analysis platform that processes uploaded resumes, extracts structured candidate data, and evaluates alignment against job descriptions.
- Developed a multi-stage evaluation pipeline for resume parsing, skill extraction, keyword matching, ATS analysis, and personalized feedback generation, enabling comprehensive resume reviews.
- Integrated LLM-based scoring and recommendation workflows to generate actionable improvement suggestions across content, formatting, experience presentation, and role-specific optimization.

CampusCart | **PTU Marketplace (Ongoing)** [GitHub]

- Built CampusCart to simplify peer-to-peer trading within college communities, implementing secure authentication, product management workflows, image uploads, and PostgreSQL-backed persistence.
- Designed and developed a buyer-seller messaging system with product-linked conversations and image sharing, enabling direct communication without relying on external platforms.
- Implemented role-based access control and protected backend APIs, ensuring secure marketplace operations and controlled access to platform resources.

ACHIEVEMENTS

- Selected for a Summer Internship at IIT Ropar (VLED Lab).
- Awarded the Merit-cum-Means Scholarship for maintaining strong academic performance during B.Tech studies.
- Recipient of the F.C. Sondhi Scholarship for ranking among the top 1% of students in school-level academics.